



- Symptom of Damage
- Identification of pathogen
- Management Strategies

Blast (*Pyricularia oryzae*)

Symptoms

- All aboveground parts of the rice plant (leaves, leaf collar, culm, culm nodes, neck, and panicle) are attacked by the fungus
- initial symptoms are white to gray-green lesions or spots with brown borders
- Small specks originate on leaves - subsequently enlarge into spindle shaped spots (0.5 to 1.5cm length, 0.3 to 0.5cm width) with ashy center.
- older lesions elliptical or spindle-shaped and whitish to gray with necrotic borders
Several spots coalesce to form big irregular patches.
- nodal infection causes the culm to break at the infected node
- Internodal infection also occurs at the base of the plant which causes white panicles similar to that induced by yellow stem borer or water deficit
- Lesions on the neck are grayish brown and causes the girdling of the neck and the panicle to fall over
- If infection of the neck occurs before milk stage, no grain is formed, but if infection occurs later, grains of poor quality are formed
- Lesions on the branches of the panicles and on the spikelet pedicels are brown to dark brown
- The size and shape of the spots vary on different rice varieties.



Lesions on leaves with grey centre and brown margin



Spindle shaped lesions on leaves



Neck Blast



Nodal Blast

Leaf Blast

- Severe cases of infection - entire crop give a blasted or burnt appearance- hence the name "BLAST"
- Severe cases - lodging of crop (after ear emergence)

Neck Blast

- Neck region of panicle develops a black color and shrivels completely / partially grain set inhibited, panicle breaks at the neck and hangs

Nodal Blast

- Nodes become black and break up

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Identification of pathogen



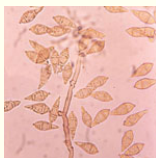
Colony of *Pyricularia oryzae*



Conidia



Conidia are pyriform (pear-like) in shape.



pyricularia oryzae microscopic

- This disease is caused by a fungus named *Pyricularia oryzae*, which overwinters in rice seeds and infected rice stubble.
- The fungus reproductive structures, spores, can spread from these two sources to rice plants during the next growing season and initiate new infections.
- Spores from these new infections can spread by wind to other rice plants over great distances.
- The conidiophores of the pathogen are produced in clusters from each stoma. They are rarely solitary with 2-4 septa. The basal area of the conidiophores is swollen and tapers toward the lighter apex.
- The conidia of the fungus measure 20-22 x 10-12 µm. The conidia are 2-septate, translucent, and slightly darkened. They are obclavate and tapering at the apex. They are truncate or extended into a short tooth at the base.
- The perfect stage is rarely found in the field.

Favourable conditions

- presence of the blast spores in the air throughout the year
- upland rice environment and high elevation in the tropics
- cloudy skies, frequent rain, and drizzles
- high nitrogen levels like ammonium sulfate
- high relative humidity (90% and higher) and wet leaves
- temperature from 25-28°C

Occurrence

- Earliest known plant disease
- Also known as rotten neck or rice fever.
- Reported from 80 rice-growing countries. First recorded in India during 1918.
- Expected grain loss : 70 to 80%

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Management Strategies

Cultural methods

- Planting resistant varieties against the rice blast is the most practical and economical way of controlling rice blast.
- Use of Tolerant varieties (**CO 47, CO 50, ADT 36, ADT 37, ASD 16, ASD 20, ADT 39, ASD 19, TPS 3, White ponni, ADT 44, BPT 5204, CORH** , Palghuna, Swarnamukhi, Swathi, Prabhat, IR - 64, , IR - 36, Jaya)
- Avoid excess N - fertilizer application
- Apply nitrogen in three split doses.
- Remove weed hosts from bunds.



Avoid Excess Use of Nitrogen Fertilizer



Use Resistant variety BPT 5204

Preventive methods

- Avoid dry nurseries.
- Avoid late planting.
- Burning of straw and stubbles after harvest
- Avoid grasses and other weeds on bunds and inside.
- Dry seed treatment with *Pseudomonas fluorescens* talc formulation @10g/kg of seed.
- Stagnate water to a depth of 2.5cm over an area of 25m² in the nursery. Sprinkle 2.5 kg of *P. fluorescens* (talc) and mix with stagnated water. Soak the root system of seedlings for 30 min and transplant.
- Spray *P. fluorescens* talc formulation @ 0.5% from 45 days after transplanting at 10 day intervals, three times.



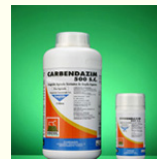
Avoid Dry Nursery



Root Dipping of Seedlings in *Pseudomonas*

Chemical methods

- Do not apply lower/higher doses of fungicides.
- Spray before 11.00 AM/after 3.00 PM.
- Avoid noon hours for spraying.
- Seed treatment at 2.0 g/kg seed with Captan or Carbendazim or Thiram or Tricyclazole.
- Systemic fungicides such as pyroquilon and tricyclazole are possible chemicals for controlling the disease.
- Spraying of Tricyclazole at 1g/lit of water or Edifenphos at 1 ml/lit of water or Carbendazim at 1.0 gm/lit.
- 3 to 4 sprays each at nursery, tillering stage and panicle emergence stage may be required for complete control.
- Nursery stage:** Light infection - Spray Carbendazim or Edifenphos @ 1.0 gm/lit.
- Pre-Tillering to Mid-Tillering:** Light at 2 to 5 % disease severities - Apply Edifenphos or Carbendazim @ 1.0 gm/lit. Delay top dressing of N fertilizers when infection is seen.
- Panicle initiation to booting:** At 2 to 5% leaf area damage- spray Edifenphos or Carbendazim or Tricyclazole @ 1.0 gm/lit.
- Flowering and after:** At 5 % leaf area damage or 1 to 2 % neck infection spray Edifenphos or Carbendazim or Tricyclazole @ 1 g /lit of water.



Spray Carbendazim



Spray Edifenphos



Thiram-Seed Treating Fungicide



Treat the Seeds with Captan

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